TASK 4 / DATA SCIENCE ASSIGNMENT

Sales Prediction Using Python

Advertising analytics and budget scenarios

TV / RADIO / NEWSPAPER

DATA AND METHODS

200 markets / Regression and scenario analysis

Source / Kaggle Advertising CSV

OVERVIEW

Executive summary

This study predicts market-level sales from spending on television, radio and newspaper advertising. It compares regression methods, evaluates adjusted channel associations and tests budget scenarios for comparable markets.

Extra Trees ranks first by training-fold RMSE (0.582 ± 0.222, versus 5.103 for the mean baseline). On the holdout it records MAE 0.381, RMSE 0.486 and R^2 0.993. Errors are in thousands of sales units.

TV has the largest absolute standardized coefficient. Newspaper has the smallest, and its HC3 95% interval (-0.014 to 0.019) includes zero. The largest holdout permutation effect belongs to TV.

Introduction and problem statement

Marketing teams need to estimate how proposed channel allocations relate to sales before committing budget. The available data are cross-sectional. The prediction question is how much a comparable market might sell under a stated budget mix.

Research objectives

Validate the original data, identify useful features, select a regression model using training folds, measure generalization on held-out markets and translate conditional predictions into testable marketing decisions.

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01 / DATA

Dataset and exploration

Advertising.csv contains 200 markets and 5 raw columns. TV, Radio and Newspaper record spend in thousands of dollars. Sales records thousands of units. The source is the Kaggle Advertising CSV distribution. The ISLR textbook documents the units, and its companion CSV matches the Kaggle data after normalizing column capitalization.

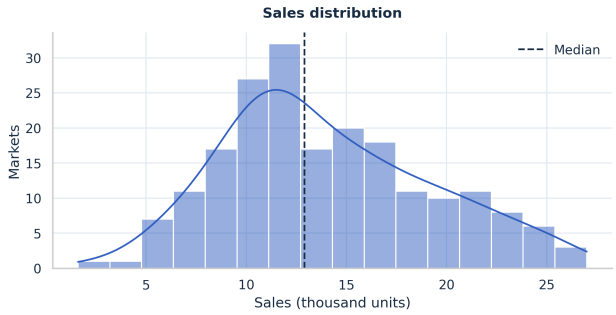
Data cleaning and preprocessing

Cleaning retained 200 of 200 markets after excluding the sequential index. It removed 0 invalid rows and 0 duplicates, and retained 2 IQR-flagged markets.

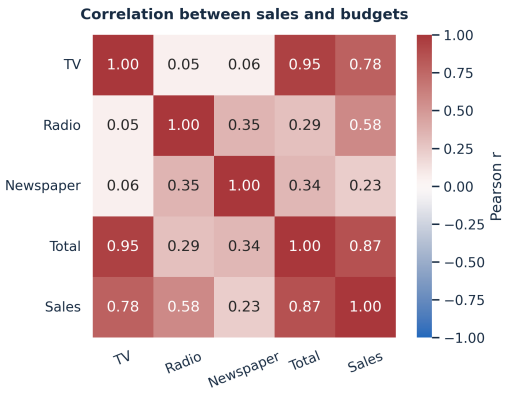
Column names were normalized, then values checked for missing, nonnumeric, infinite and negative entries. The IQR flag identifies observations for review. Original records and a separate cleaning audit were preserved.

Field	Mean	Median	SD	Min	Max	Skew
TV	147.04	149.75	85.85	0.70	296.40	-0.07
Radio	23.26	22.90	14.85	0.00	49.60	+0.09
Newspaper	30.55	25.75	21.78	0.30	114.00	+0.89
Total	200.86	207.35	92.99	11.70	433.60	+0.05
Sales	14.02	12.90	5.22	1.60	27.00	+0.41

Distributions and relationships



Sales averages 14.02 with a median of 12.90. Sample skewness is +0.41. The dashed line marks the median.



TV has r = 0.78 with sales and total spend has r = 0.87. Total inherits information from its component budgets.

The middle half of sales spans 10.38 to 17.40 thousand units. The sales Shapiro-Wilk p-value is 0.00168. Predictor distributions need not be normal for OLS. The descriptive CSV provides skewness, excess kurtosis, quartiles and normality checks for all numeric variables.

Feature and model selection

The predictors were split randomly into 160 training markets and 40 holdout markets using seed 42. All candidates used identical 5-fold cross-validation repeated five times. Feature transforms and any scaling were inside each pipeline.

Feature engineering and selection

Candidate sets compare raw channels, total spend, total with two channel shares, TV and Radio alone, and selected interaction and quadratic terms. Total was kept separate from its three components to avoid exact collinearity. Mean validation RMSE ranks the models. Compact parameter grids tune regularized and ensemble models.

Candidate	CV RMSE	CV SD	Test MAE	Test RMSE	Test R^2
Extra Trees	0.582	0.222	0.381	0.486	0.993
Quadratic TV and Radio OLS	0.614	0.267	0.493	0.603	0.988
Quadratic plus Newspaper OLS	0.624	0.265	0.492	0.607	0.988
Gradient Boosting	0.798	0.178	0.546	0.652	0.987
Random Forest	0.865	0.206	0.613	0.756	0.982
TV and Radio interaction OLS	0.943	0.309	0.666	0.902	0.974
TV and Radio OLS	1.649	0.351	1.444	1.771	0.901
Lasso Regression	1.662	0.345	1.461	1.791	0.898
ElasticNet Regression	1.663	0.348	1.461	1.784	0.899
Multiple Linear Regression	1.664	0.348	1.461	1.782	0.899
Ridge Regression	1.664	0.348	1.461	1.782	0.899
Total and mix OLS	2.481	0.441	2.044	2.660	0.776
Total spend OLS	2.550	0.253	2.088	2.789	0.754
Simple Linear Regression	3.285	0.240	2.444	3.194	0.677
Dummy Mean Baseline	5.103	0.515	4.963	5.631	-0.005

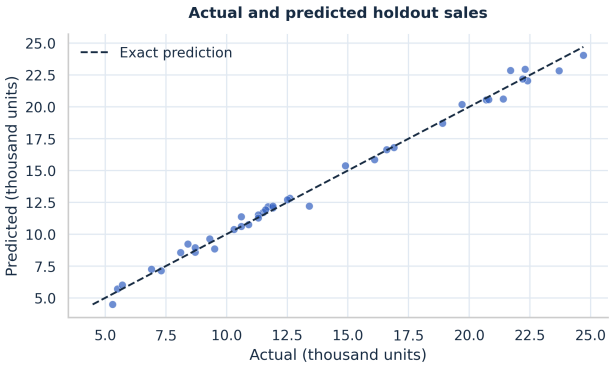
Selected inputs and derived terms include TV, Radio. The ranking uses training CV. Holdout results are shown for all frozen candidates for comparison.

Adding the TV × Radio interaction changes mean validation RMSE from 1.649 to 0.943. The additive three-channel OLS records 1.664. The interaction tests whether a channel’s predictive relationship varies with the level of the other channel.

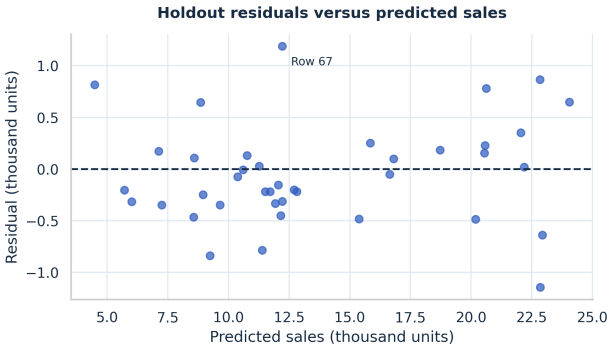
Holdout model performance

Extra Trees ranks first by training-fold RMSE (0.582 ± 0.222, versus 5.103 for the mean baseline). On the holdout it records MAE 0.381, RMSE 0.486 and R^2 0.993. Errors are in thousands of sales units.

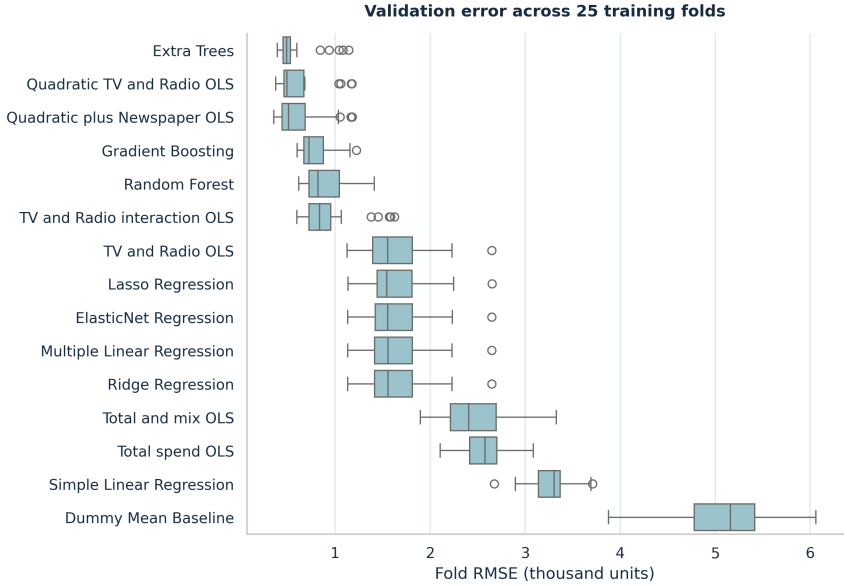
MAE is the average absolute miss. MSE averages squared misses, and its square root gives RMSE in the original sales scale. R^2 describes squared-error improvement over predicting the evaluation-set mean. Adjusted R^2 is reported for training OLS fits and penalizes added terms.



Points close to the diagonal are better predictions. Points above it are overpredictions.



Mean holdout residual is -0.047 thousand units. Spearman correlation of residual with prediction is +0.16 (p = 0.333). The largest absolute miss is 1.19 at source row 67 (zero-based). Winner train-fold RMSE averages 0.001 versus 0.582 on validation folds. The near-zero training error shows the flexibility of the fit. Holdout RMSE (0.486) is close to validation RMSE, and the residuals show little monotone trend.



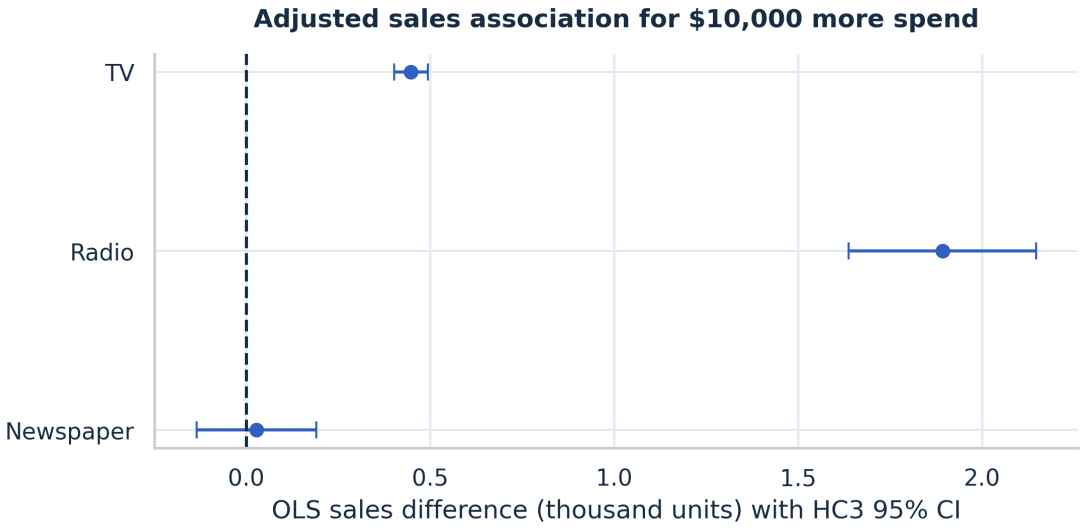
Across the shared folds, the winner averages 0.582 RMSE with fold SD 0.222. Fold SD describes stability across overlapping splits.

Adjusted channel associations

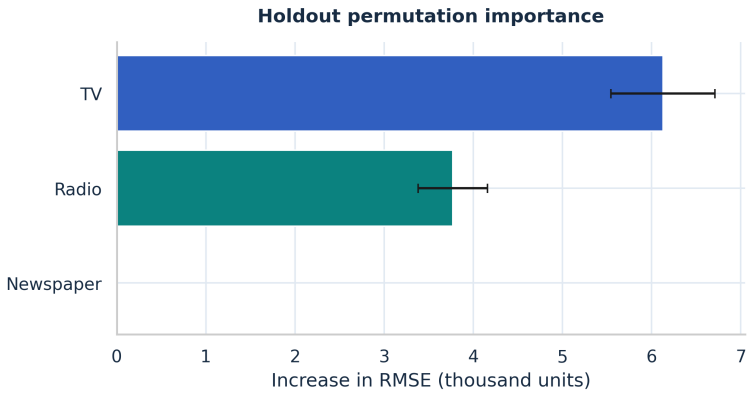
A separate OLS model uses the three original channel budgets and training markets. Each slope estimates the sales difference associated with one additional thousand dollars in a channel, holding the other budgets fixed. HC3 standard errors address unequal residual variance.

Term	Slope	HC3 SE	p-value	95% interval	VIF
Intercept	2.9791	0.3969	6.1e-14	2.2011 to 3.7570	—
TV	0.0447	0.0023	5.5e-82	0.0402 to 0.0493	1.00
Radio	0.1892	0.0130	9.1e-48	0.1637 to 0.2147	1.18
Newspaper	0.0028	0.0083	0.74	-0.0136 to 0.0191	1.18

Training OLS R^2 is 0.896, adjusted R^2 is 0.894, and robust F is 474.5 ($p = 3.6e-78$). The highest VIF is 1.18, suggesting little variance inflation in this specification.



TV has the largest absolute standardized coefficient. Newspaper has the smallest, and its HC3 95% interval (-0.014 to 0.019) includes zero. The largest holdout permutation effect belongs to TV.



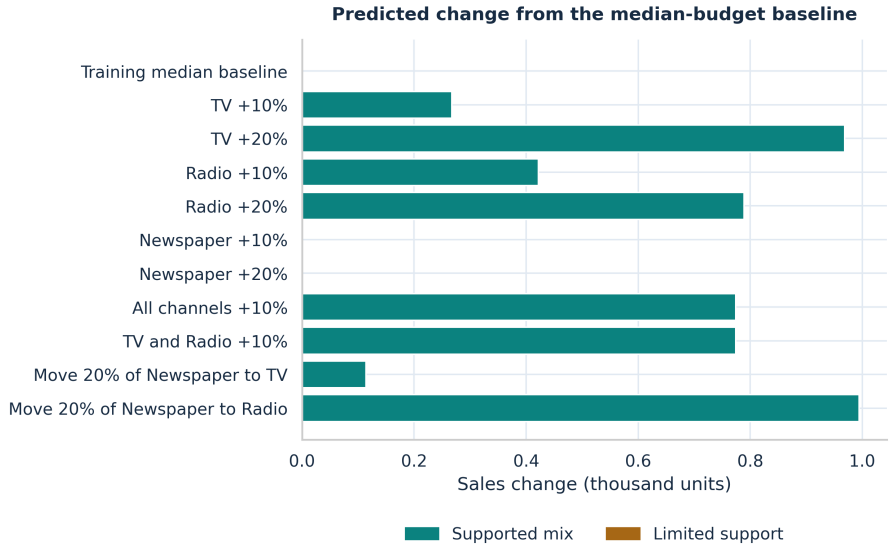
Shuffling TV produces the largest mean increase in holdout RMSE. Error bars show shuffle-to-shuffle SD.

Sales at proposed budgets

The baseline uses the training median of each budget. The saved model scores the higher-spend and constant-budget alternatives. Spend is in thousands of dollars, and sales and changes are in thousands of units.

Scenario	Total spend	Pred. sales	Sales change	Support
Training median baseline	197.45	14.37	+0.00	Within
TV +10%	212.52	14.64	+0.27	Within
TV +20%	227.58	15.34	+0.97	Within
Radio +10%	199.57	14.80	+0.42	Within
Radio +20%	201.69	15.16	+0.79	Within
Newspaper +10%	200.01	14.37	+0.00	Within
Newspaper +20%	202.57	14.37	+0.00	Within
All channels +10%	217.20	15.15	+0.77	Within
TV and Radio +10%	214.64	15.15	+0.77	Within
Move 20% of Newspaper to TV	197.45	14.49	+0.11	Within
Move 20% of Newspaper to Radio	197.45	15.37	+0.99	Within

The simulator checks each channel’s training min and max. It also compares the proposal’s nearest training mix with the 95th percentile of training nearest-neighbor distances, after scaling channels by their training ranges.



Among supported examples, Move 20% of Newspaper to Radio has the largest predicted change (+0.99 thousand units).

Recommendations

01 Prioritize TV in controlled budget tests and vary TV and Radio jointly. The interaction model reduced validation error relative to the additive model.

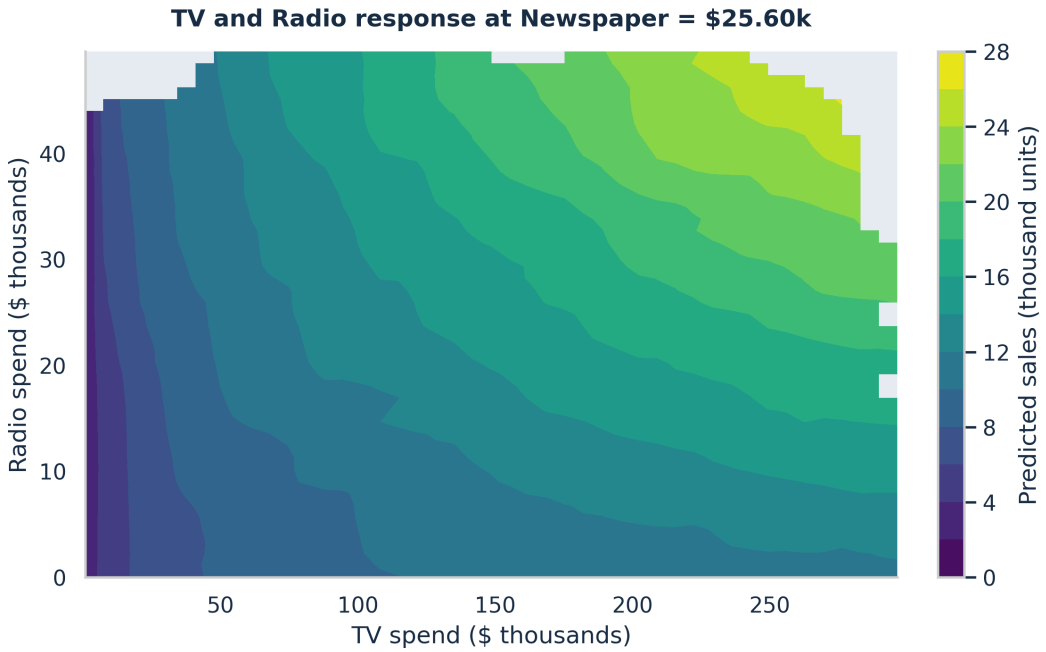
02 Audit Newspaper against reach, cost and campaign objectives. The selected model uses TV, Radio, so varying Newspaper alone leaves its predictions unchanged.

03 At the same total budget, the scenario “Move 20% of Newspaper to Radio” yields a predicted gain of 0.99 thousand units. Pilot the reallocation while tracking incremental sales.

04 Among the tested spend increases, TV +20% produces the largest supported gain (+0.97 thousand units). Compare the cost and incremental sales in a controlled test.

05 Capture campaign dates, prices, promotions and audience segments so future models can test seasonality and segment response.

TV and Radio response surface



Newspaper is fixed at the training median ($25.60 thousand). Gray indicates sparse combinations. 96.1% of grid points meet the training support rule.

07 / INTERPRETATION

Limitations and conclusion

The file covers 200 historical markets and has no calendar or segment identifiers. Estimates apply to budget scenarios for comparable markets.

Market-level associations can reflect omitted variables such as pricing, promotions or market size. Use randomized tests to estimate incremental effects.

Scenario predictions are point estimates. The range and nearest-neighbor flags measure coverage of the training data and provide no prediction interval.

Repeated CV folds overlap, and tuning and ranking share those folds. The holdout provides the final evaluation and contains only 40 markets.

Conclusion

The holdout results quantify how closely advertising budgets predict sales in this sample. The scenario simulator translates those relationships into comparable budget options and flags unfamiliar mixes. Controlled marketing tests can establish whether the strongest predicted changes translate into incremental sales.

References

Bumba5341. Advertising.csv. Kaggle. Dataset source.

James, Witten, Hastie and Tibshirani. An Introduction to Statistical Learning, 2013. Figure 2.1 and Chapter 3.

Scikit-learn user guide. Cross-validation and model evaluation.

Statsmodels documentation. Ordinary least squares and robust covariance.

The raw CSV was retrieved on 2026-11-09. File size and SHA-256 are recorded in data/raw/dataset_metadata.json.